

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	XXXXXX
Project Name	Credit Card Auto-Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Interface	Web form fields that collect demographic, financial, and employment inputs from the user.	predict.html (UI)	Done	Core / High
2	Machine Learning Pipeline	Fully preprocessed classifier model loaded via pickle file for high-accuracy predictions.	final_credit_model_pipeline.pkl	Done	Core / High
3	Real-Time Prediction Engine	Backend route that processes POST requests, feeds input details into the model, and calculates results.	app.py (Backend)	Done	Core / High
4	Dynamic Results Visualizer	Displays prediction output (Approved/Rejected) and confidence probability percentages on a styled card.	result.html (UI)	Done	Core / Medium
5	Automated EDA Visualizations	Interactive plots mapped to static directory showcasing data properties like income and age distributions.	static/img/ (Visuals)	Done	Core / Medium
6	Dynamic Port Management	Automatic switching of routing ports (e.g., to 8080) if standard ports are occupied.	app.py (Core Logic)	Done	Nice-to-Have / Low
7	Project Home Page	Dynamic index page introducing key aspects, accuracy, and workflow of the classifier system.	index.html (UI)	Done	Core / Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	7(Input UI, ML Pipeline, Prediction Engine, Results Card, EDA Visuals, Home page, Navigation Layout)
Additional / Nice-to-Have Features	1Dynamic Port Management / Fallback Configuration)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	• Landing Page (index.html) • Form Inputs (predict.html) • Results Display Page (result.html) • Base Master Template (base.html)
2	Backend / Logic	2	• Real-time Prediction Route (app.py) • Dynamic Server Port Fallback Check
3	Database / Storage	1	• ML Model Pipeline Pickle File Serialization (final_credit_model_pipeline.pkl)
4	API / Integration	1	• Data Integration passing client-side HTML form data to backend scikit-learn models
5	Security / Authentication	0	• Input sanitization and field-level validation on the HTML input fields